TEXORA · LECTURE SHEET

Project Plan for Knitting Shed

A knitting shed planned the way Sheet 6 planned a weaving shed: a monthly tonnage becomes machines, machines become floor, power and people, and every figure is checked against the rates the course has already verified. The machines here are Sheet 12's — the plan is deliberately built from numbers already proven.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 14 · SPECIALIZATION — FABRIC MANUFACTURING

1. The project brief

ITEM	VALUE
Product	Cotton single jersey, 179 GSM, 176 cm open width (Sheet 12's fabric)
Target output	60 tonnes per month, sustained
Machine	26 in, 28 G, 72 feeders, 18 rpm — plan rate **101.7 kg per shift** at 85% (Sheet 12)
Working pattern	3 shifts/day, 26 days/month = 78 shifts per machine per month

2. Step 1 — Machines

WORKED CALCULATION — FLEET SIZE

Output per machine per month = 101.7 × 78 = **7,933 kg**.

Machines = 60,000 ÷ 7,933 = 7.56 → **8 machines**.

Installed capacity = 8 × 7,933 = **63,461 kg/month** — a 5.8% margin over target. That margin is thinner than Sheet 12's single-order margin, deliberately so: a shed is a permanent asset, and its margin is managed by scheduling (a ninth machine's space reserved in the layout, Section 4) rather than by buying capacity that stands idle in year one.

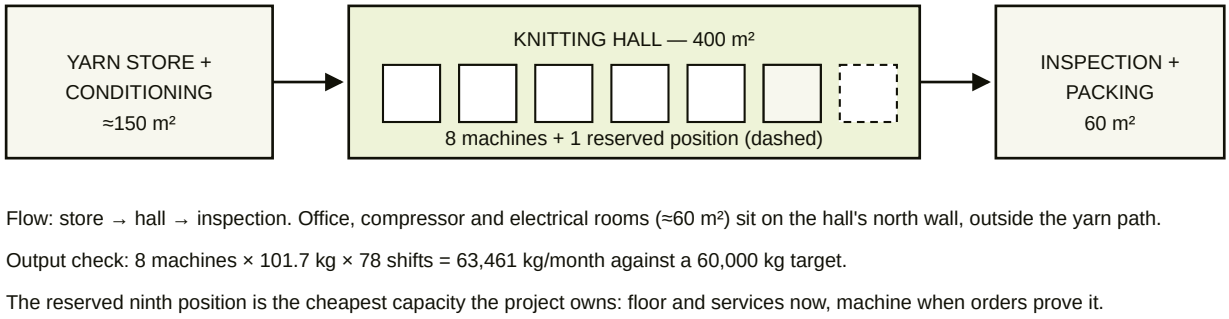
3. Step 2 — Yarn store and winding

- **Yarn throughput:** 60,000 × 1.02 = **61,200 kg/month** of Ne 24 (Sheet 10's allowance) — 2,354 kg per working day arriving, being conditioned and being issued.

- **Store sizing:** two weeks' buffer = ≈30,600 kg. At Sheet 1's cone weight of 1.36 kg that is about 22,500 cones in store at any time — the figure that sizes the racks, the conditioning room and the forklift aisle, none of which appear in the knitting hall at all.
- **Cone handling:** 2,354 kg/day ÷ 1.36 kg ≈ 1,730 cones/day creelled — roughly one cone change every 50 seconds somewhere in the shed, around the clock; the labour plan must contain this number or the machines will wait for yarn with full orders on the wall.

4. Step 3 — Floor and layout

Installed area per circular machine, including its working apron and share of aisles, is about 12 m²; the knitting hall for 8 machines plus the reserved ninth position, inspection tables and a fabric buffer bay is planned at **400 m²**. Support spaces — yarn store and conditioning (≈150 m²), inspection/packing (60 m²), office, compressor and electrical rooms (60 m²) — bring the project to ≈670 m² under one roof, arranged so yarn enters at one end and fabric leaves at the other.



Flow: store → hall → inspection. Office, compressor and electrical rooms (≈60 m²) sit on the hall's north wall, outside the yarn path.

Output check: 8 machines × 101.7 kg × 78 shifts = 63,461 kg/month against a 60,000 kg target.

The reserved ninth position is the cheapest capacity the project owns: floor and services now, machine when orders prove it.

Figure 1 — Knitting shed layout: one-way flow from yarn store to packing, with growth built into the plan.

5. Step 4 — Power and people

- **Power:** machine motors 8 × 5.5 kW = 44 kW; auxiliaries (lighting, air conditioning of the hall, compressor, inspection) at +30% → connected load ≈ **57 kW**. Knitting is the most power-frugal fabric process per kilogram in the mill — compare the weaving shed's lighting-and-drive load per metre before believing any "textiles are power-hungry" generalisation.
- **People per shift:** 2 knitters (4 machines each), 1 helper/material handler (the 1,730 cones/day of Section 3 are his arithmetic), 1 mechanic shared with the next shed, 1 supervisor per day across shifts, plus inspection staff scaled to metres, not machines.
- **Quality station:** GSM and width checked per roll against Sheet 12's closing check — the shed's plan rate means nothing if its cloth is off-specification, so the plan staffs the check, not just the machines.

6. The plan on one line

8 machines × 101.7 kg/shift × 78 shifts = 63,461 kg/month ≥ 60,000 kg target; 61,200 kg of yarn a month through a two-week store; ≈670 m² of building; ≈57 kW connected; a ninth machine's floor reserved. Every number in that sentence was calculated earlier in this course — a project plan is not new mathematics, it is old mathematics that agrees with itself.

Glossary

Shed plan — the complete capacity, space, power and staffing calculation for a production unit. **Buffer stock** — yarn held to decouple delivery rhythm from consumption rhythm. **Connected load** — total installed electrical demand. **Reserved position** — floor and services provided for a future machine.

Key points

• Fleet = target ÷ (rate × shifts): 60,000 ÷ (101.7 × 78) → 8 machines, 63,461 kg installed.

• Yarn system sized from the same plan: 61,200 kg/month, ≈30,600 kg buffer, ≈1,730 cones creelled per day.

• Space ≈670 m² total, flow one-way; growth reserved as floor, not bought as a machine.

• Power ≈57 kW connected — knitting is frugal per kg; people follow machines and metres, in that order.

Practice

1. The target rises to 75 tonnes/month. How many machines are required, and does the reserved position suffice?
2. Why is the shed's margin (5.8%) acceptable here when Sheet 12 wanted 14% on a single order?
3. Recompute the daily cone-creeling figure if cone weight rises to 2.0 kg.

Answer key

1. 75,000 ÷ 7,933 = 9.45 → **10 machines**: the reserved position covers the ninth, but a tenth needs new floor — the plan's growth path ends exactly where its layout said it would.
2. A single order has no tomorrow: miss it and the capacity is stranded. A shed re-plans every month — scheduling, overtime shifts and the reserved position are all live levers — so it can run a thinner standing margin and buy capacity only when the order book proves it.
3. 2,354 ÷ 2.0 = **1,177 cones/day** — heavier cones mean fewer creel changes and less handling labour for the same yarn weight; package size is a labour variable as much as a store variable (Sheet 1).